



DIRECTIONS FOR USE

B. Braun Melsungen AG - Carl-Braun-Straße 1, 34212 Melsungen, Germany

Aminoplasma® B. Braun 10% E
Solution for Intravenous Infusion

1. NAME OF THE MEDICINAL PRODUCT

Aminoplasma® B. Braun 10% E Solution for Intravenous Infusion

2. QUALITATIVE AND QUANTITATIVE COMPOSITION

The solution for infusion contains:

	per 1 ml	per 250 ml	per 500 ml	per 1000 ml
Isoleucine	5.00 mg	1.25 g	2.50 g	5.00 g
Leucine	8.90 mg	2.23 g	4.45 g	8.90 g
Lysine hydrochloride	8.56 mg	2.14 g	4.28 g	8.56 g
(equivalent to lysine)	(6.85 mg)	(1.71 g)	(3.43 g)	(6.85 g)
Methionine	4.40 mg	1.10 g	2.20 g	4.40 g
Phenylalanine	4.70 mg	1.18 g	2.35 g	4.70 g
Threonine	4.20 mg	1.05 g	2.10 g	4.20 g
Tryptophan	1.60 mg	0.40 g	0.80 g	1.60 g
Valine	6.20 mg	1.55 g	3.10 g	6.20 g
Arginine	11.50 mg	2.88 g	5.75 g	11.50 g
Histidine	3.00 mg	0.75 g	1.50 g	3.00 g
Alanine	10.50 mg	2.63 g	5.25 g	10.50 g
Glycine	12.00 mg	3.00 g	6.00 g	12.00 g
Aspartic acid	5.60 mg	1.40 g	2.80 g	5.60 g
Glutamic acid	7.20 mg	1.80 g	3.60 g	7.20 g
Proline	5.50 mg	1.38 g	2.75 g	5.50 g
Serine	2.30 mg	0.58 g	1.15 g	2.30 g
Tyrosine	0.40 mg	0.10 g	0.20 g	0.40 g
Sodium acetate trihydrate	2.858 mg	0.715 g	1.429 g	2.858 g
Potassium acetate	2.453 mg	0.613 g	1.227 g	2.453 g
Sodium hydroxide	0.360 mg	0.090 g	0.180 g	0.360 g
Magnesium chloride hexahydrate	0.508 mg	0.127 g	0.254 g	0.508 g
Disodium phosphate dodecahydrate	3.581 mg	0.895 g	1.791 g	3.581 g

Electrolyte concentrations

Sodium	50	mmol/l
Potassium	25	mmol/l
Magnesium	2.5	mmol/l
Acetate	46	mmol/l
Chloride	52	mmol/l
Phosphate	10	mmol/l
Citrate	1.0 – 2.0	mmol/l
Total amino acids	100	g/l
Total nitrogen	15.8	g/l

3. PHARMACEUTICAL FORM
Solution for infusion

Clear, colourless up to faintly straw-coloured aqueous solution	
Energy [kJ/l (kcal/l)]	1675 (400)
Theoretical osmolarity [mOsm/l]	1021
Acidity (titration to pH 7.4) [mmol NaOH/l]	approx. 26
pH	5.7 – 6.3

4. CLINICAL PARTICULARS

4.1 Therapeutic indications

Supply of amino acids and a limited amount of electrolytes for par-
enteral nutrition, when oral or enteral nutrition is impossible, insuffi-
cient or contraindicated. For adults, adolescents and children over
2 years of age.

4.2 Posology and method of administration

Posology

The dosage has to be adjusted according to the individual need of
amino acids, electrolytes, and fluid depending on the clinical condition
of the patient (nutritional status and/or degree of nitrogen catabolism
due to underlying disease).

Adults and adolescents from 14 to 17 years

Daily dose:

1.0 – 2.0 g amino acids/kg body weight $\pm$ 10 – 20 ml/kg body weight
 $\pm$ 700 – 1400 ml for a 70 kg patient

Maximum infusion rate:

0.1 g amino acids/kg body weight/h $\pm$ 1.0 ml/kg body weight/h
 $\pm$ 1.17 ml/min for a 70 kg patient

Paediatric population

Newborn infants, infants and toddlers less than two years of age

Aminoplasma® B. Braun 10% E is contraindicated in newborn infants,
infants and toddlers less than 2 years of age (see section 4.3).

Children and adolescents 2 to 13 years

The dosages for the age groups stated below are average values for
guidance. The exact dosage should be adjusted individually according
to age, developmental stage and prevailing disease.

Daily dose for children 2 to 4 years old:

1.5 g amino acids/kg body weight $\pm$ 15 ml/kg body weight

Daily dose for children 5 to 13 years old:

1.0 g amino acids/kg body weight $\pm$ 10 ml/kg body weight

Critically ill children: For critically ill patients the advisable amino acid
intake may be higher (up to 3.0 g amino acids/kg body weight per
day).

Maximum infusion rate:

0.1 g amino acids/kg body weight/h $\pm$ 1 ml/kg body weight/h

In the case of amino acid requirements of 1.0 g/kg body weight/day or
more, particular attention should be paid to the limitations of fluid
input. To avoid fluid overload, amino acid solutions with higher amino
acid content may have to be used in such situations.

Patients with renal/hepatic impairment

The doses should be adjusted individually in patients with hepatic or
renal insufficiency (see also section 4.4). Aminoplasma® B. Braun 10% E
is contraindicated in severe hepatic insufficiency and severe renal
insufficiency in absence of renal replacement therapy (see section
4.3).

Duration of use

This solution can be administered as long as parenteral nutrition is
indicated.

Method of administration

Intravenous use.

For central venous infusion only.

4.3 Contraindications

- Hypersensitivity to any of the active substances or to any of the
excipients listed in section 6.1
- Inborn errors of amino acid metabolism
- Severe circulation disorders with vital risk (e.g. shock)
- Hypoxia
- Metabolic acidosis
- Severe hepatic insufficiency
- Severe renal insufficiency in absence of renal replacement therapy
- High and uncorrected plasma concentration of one of the elec-
trolytes contained in the product
- Decompensated cardiac insufficiency
- Acute pulmonary oedema
- Disturbances of the electrolyte and fluid balance

The medicinal product must not be administered to newborn infants,
infants and toddlers less than two years of age, because the amino
acid composition does not properly meet the special requirements of
this paediatric age group.

4.4 Special warnings and precautions for use

The medicinal product should only be administered after careful ben-
efit-risk assessment in the presence of disorders of amino acid metab-
olism of other origin than stated under section 4.3.

Care should be exercised in the administration of large volume infu-
sion fluids to patients with cardiac insufficiency.

Caution should be exercised in patients with increased serum osmo-
larity.

Disturbances of fluid and electrolyte balance (e.g. hypotonic dehydra-
tion, hyponatraemia, hypokalaemia) should be corrected prior to the
administration of parenteral nutrition.

Serum electrolytes, blood glucose, fluid balance, acid-base balance
and renal function should be monitored regularly.

Monitoring should also include serum protein and liver function tests.

In patients with renal insufficiency, the dose must be carefully adjust-
ed according to individual needs, severity of organ insufficiency and
the kind of instituted renal replacement therapy (haemodialysis,
haemofiltration etc.).

In patients with hepatic insufficiency, the dose must be carefully
adjusted according to individual needs and severity of organ insuffi-
ciency.

Amino acid solutions are only one component of parenteral nutrition.
For complete parenteral nutrition, substrates for non-protein energy
supply, essential fatty acids, electrolytes, vitamins, fluids and trace
elements must be administered together with amino acids.

4.5 Interaction with other medicinal products and other forms of
interaction

None known.

4.6 Fertility, pregnancy and lactation

Pregnancy

There are no or limited amount of data from the use of Aminoplasma®
B. Braun 10% E in pregnant women. The use of Aminoplasma®
B. Braun 10% E may be considered during pregnancy, if necessary.

Breastfeeding

Amino acids/metabolites are excreted in human milk, but at thera-
peutic doses of Aminoplasma® B. Braun 10% E no effects on the
breastfed newborns/infants are anticipated.

Nevertheless, breast-feeding is not recommended for mothers on par-
enteral nutrition.

Fertility

No data available.

4.7 Effects on ability to drive and use machines

Not relevant.

4.8 Undesirable effects

Undesirable effects that, however, are not specifically related to the
product but to parenteral nutrition in general may occur, especially at
the beginning of parenteral nutrition.

Undesirable effects are listed according to their frequencies as fol-
lows:

- Very common ($\geq 1/10$)
- Common ($\geq 1/100$ to $< 1/10$)
- Uncommon ($\geq 1/1,000$ to $< 1/100$)
- Rare ($\geq 1/10,000$ to $< 1/1,000$)
- Very rare ($< 1/10,000$)
- Not known (cannot be estimated from the available data)

Immune system disorders

Not known: Allergic reactions

Gastrointestinal disorders

Uncommon: Nausea, vomiting

Note:

Patients should inform their doctor or pharmacist if they notice any
side effect not mentioned in this leaflet.

4.9 Overdose

Symptoms of fluid and electrolytes overdose

Overdose or too high infusion rates may lead to hyperhydration, elec-
trolyte imbalance and pulmonary oedema.

Symptoms of amino acid overdose

Overdose or too high infusion rates may lead to intolerance reactions
manifesting in the form of sickness, vomiting, shivering, headache,
metabolic acidosis, hyperammonaemia and renal amino acid losses.

Treatment

If intolerance reactions occur, the amino acid infusion must be inter-
rupted temporarily and resumed later on at a lower infusion rate

5. PHARMACOLOGICAL PROPERTIES

5.1 Pharmacodynamic properties

Pharmacotherapeutic group: Blood substitutes and perfusion solu-
tions, i.v. solutions for parenteral nutrition, combinations.

ATC code: B05BA10

Mechanism of action

The aim of parenteral nutrition is the supply of all nutrients necessary
for the growth, maintenance and regeneration of body tissues etc.

Approval for Printing
B. BRAUN Melsungen AG

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New draft required ☐

LRA:

Date Signature

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Amino acids are of special importance as they partly are essential for protein synthesis. Intravenously administered amino acids are incorporated in the respective intravascular and intracellular amino acid pools. Both endogenous and exogenous amino acids serve as substrate for the synthesis of functional and structural proteins.

Electrolytes administered with parenteral nutrition help to maintain the serum levels necessary for the physiological processes of the cell.

To prevent the metabolisation of amino acids for energy production, and also to fuel the other energy-consuming processes in the organism, simultaneous non-protein energy supply (in the form of carbohydrates or fats) is necessary.

5.2 Pharmacokinetic properties

Absorption

Because this medicinal product is infused intravenously, the bio-availability of the amino acids contained in the solution is 100%.

Distribution

Amino acids are incorporated in a variety of proteins in different tissues of the body. In addition each amino acid is present as free amino acid in the blood and inside cells.

The composition of the amino acid solution is based upon the results of clinical investigations of the metabolism of intravenously administered amino acids. The quantities of the amino acids contained in the solution have been chosen so that a homogenous increase of the concentrations of all plasma amino acids is achieved. The physiological ratios of plasma amino acids, i.e. the amino acid homeostasis, are thus maintained during infusion of the medicinal product.

Normal foetal growth and development depend on a continuous supply of amino acids from the mother to the foetus. The placenta is responsible for the transfer of amino acids between the two circulations.

Biotransformation

Amino acids that do not enter protein synthesis are metabolised as follows: The amino group is separated from the carbon skeleton by transamination. The carbon chain is either oxidised directly to CO₂ or utilised as substrate for gluconeogenesis in the liver. The amino group is also metabolised in the liver to urea.

Elimination

Only minor amounts of amino acids are excreted unchanged in the urine.

5.3 Preclinical safety data

Non-clinical data available for the single components of the medicinal product reveal at common dosages no special hazard for humans based on conventional data of safety pharmacology, repeated dose toxicity, genotoxicity, carcinogenic potential, toxicity to reproduction and development.

Therefore, no toxic reactions are expected to occur as long as the indications, contraindications and dosage recommendations are duly observed.

6. PHARMACEUTICAL PARTICULARS

6.1 List of excipients

Acetylcysteine

Citric acid monohydrate (for pH-adjustment)

Water for injections

6.2 Incompatibilities

Aminoplasma[®] B. Braun 10% E can only be mixed with other nutrients such as carbohydrates, lipids, vitamins and trace elements for which compatibility has been documented.

Compatibility data for different additives (e.g. electrolytes, trace elements, vitamins) and the corresponding shelf life of such admixtures can be provided on demand by the manufacturer. See also section 6.6.

6.3 Shelf life

Unopened

24 months

After first opening

The medicinal product should be used immediately.

After admixture of additives

From a microbiological point of view, mixtures should be administered immediately after preparation. If not administered immediately, storage times and conditions of mixtures prior to use are the responsibility of the user and would normally not be longer than 24 hours at 2°C – 8°C, unless mixing has taken place under controlled and validated aseptic conditions.

6.4 Special precautions for storage

Do not store above 30°C.

Keep the bottle in the outer carton in order to protect from light.

Cool storage of the solution, below 15 °C, may lead to formation of crystals, that can, however, be easily dissolved by gentle warming at 25 °C until dissolution is complete. Shake container gently to ensure homogeneity.

Do not freeze.

6.5 Nature and contents of container

Bottles of colourless glass (type II), sealed with halogen butyl rubber stoppers, containing 500 ml of solution.

Pack size: 10 x 500 ml

6.6 Special precautions for disposal and other handling

No special requirements for disposal.

Containers are for single use only. Discard container and any unused contents after use.

Only to be used if the solution is clear and colourless up to faintly straw-coloured and the bottle and its closure are undamaged.

Use a sterile giving set for administration.

If in the setting of complete parenteral nutrition it is necessary to add other nutrients such as carbohydrates, lipids, vitamins, electrolytes and trace elements to this medicinal product, admixing must be performed under strict aseptic conditions. Mix well after admixture of any additive. Pay special attention to compatibility.

Keep out of the reach and sight of children.

7. DATE OF REVISION OF THE TEXT

12.2020



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